

Optimización tecno-económica de proyectos de autogeneración AGPE con PV y BESS



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Ciclo de 6 semanas – 12 sesiones

Módulo 2 – Clase 3 - 12 de mayo de 2026



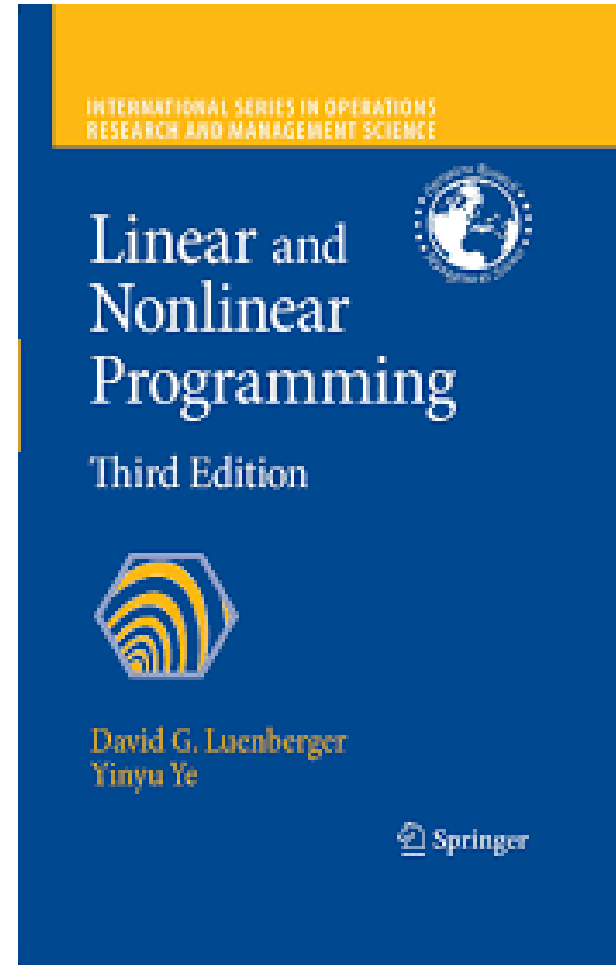
Agenda

1. Introducción a la optimización matemática en Mercados Eléctricos (Continuación)
2. Despacho económico centralizado óptimo
3. Construyendo modelos de despacho de AGPE

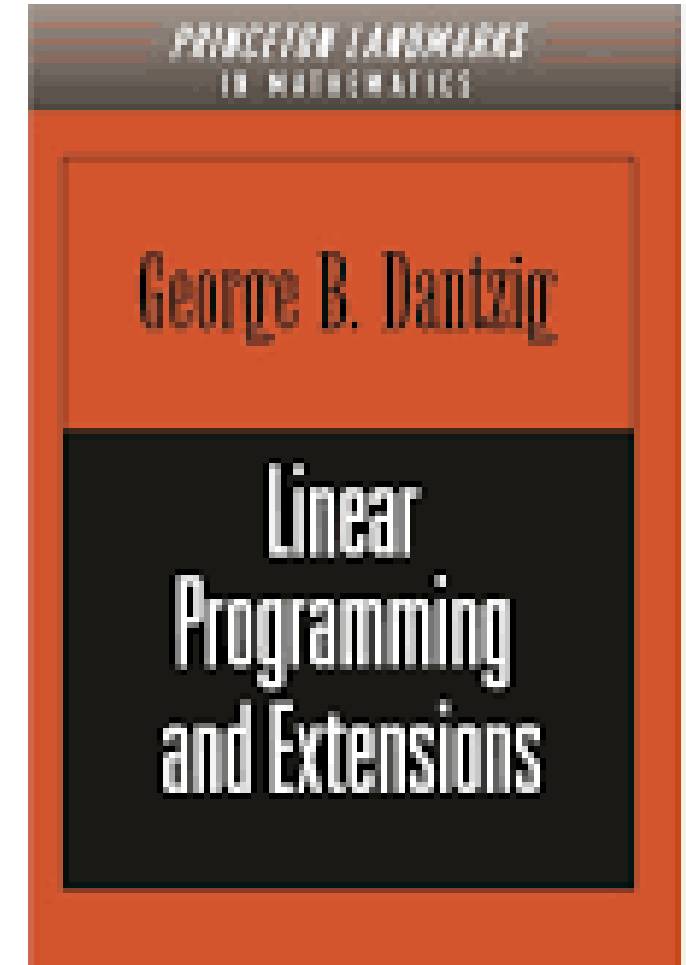
Investigación de Operaciones



Hiller & Lieberman



Luemberg



Dantzig

1. Introducción a la Optimización Matemática

En este módulo tiene como objetivo conocer o repasar los modelos matemáticos de optimización matemática requeridos para la gestión integral de proyectos energéticos sostenibles.

Se estudiarán dos formulaciones que dependiendo de su naturaleza podrán ser lineales (LP) o no lineales-cuadráticos (NLP):

- Programación Lineal LP
- Programación No Lineal Cuadrática NLP/QP

El repositorio de software utilizado en esta guía de estudio se encuentra en la siguiente página [GitHub](#).

https://github.com/pmdeoliveiradejesus/AGyCE/tree/main/M1_Optimization

El problema primal establecido en las Ecs. 1-4 pueden escribirse de forma compacta (matricial) de la siguiente forma:

$$\begin{aligned} &\text{minimizar} && \mathbf{c}^T \mathbf{x} \\ &\text{sujeto a} && \mathbf{Ax} \leq \mathbf{b} \\ &&& \mathbf{x} \geq \mathbf{0} \end{aligned}$$

La correspondiente formulación dual se escribe de la siguiente forma:

$$\begin{aligned} &\text{maximizar} && \boldsymbol{\lambda}^T \mathbf{b} \\ &\text{sujeto a} && \boldsymbol{\lambda}^T \mathbf{A} \leq \mathbf{c}^T \\ &&& \boldsymbol{\lambda} \leq \mathbf{0} \end{aligned}$$

2.1. Ejemplo 1 - LP

A continuación se presenta un ejemplo ilustrativo de la solución de un problema LP:

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3$$

Sujeto a:

$$x_1 + 2x_2 + x_3 \leq 10,$$

$$2x_1 + x_2 + 3x_3 \leq 15,$$

$$x_1, x_2, x_3 \geq 0.$$

Este problema está subdeterminado ($n > m$) por cuanto tiene tres incógnitas ($n=3$) con solo dos restricciones ($m=2$). Existen infinitas soluciones que satisfacen ambas restricciones, pero solo una solución que cumple con el objetivo establecido.

Solución óptima primal es $(x_1, x_2, x_3) = (6.667, 1.6667, 0)$ con $Z = 28.33$.

Resultados en: https://github.com/pmdeoliveiradejesus/AGyCE/tree/main/M1_Optimization

Haciendo pruebas con una hoja de Excel

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3 \quad \max \quad Z = 3 \boxed{1} + 5 \boxed{1} + 2 \boxed{1} = 10$$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Sujeto a:

$$\begin{aligned} 1 \quad \boxed{1} + 2 \quad \boxed{1} + 1 \quad \boxed{1} &= 4 <= 10 \\ 2 \quad \boxed{1} + 1 \quad \boxed{1} + 3 \quad \boxed{1} &= 6 <= 15 \end{aligned}$$

Ejemplo ilustrativo:

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3 \quad \max \quad Z = 3 \boxed{4} + 5 \boxed{3} + 2 \boxed{1} = 29$$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Sujeto a:

$$\begin{aligned} 1 \quad \boxed{4} + 2 \quad \boxed{3} + 1 \quad \boxed{1} &= 11 <= 10 \\ 2 \quad \boxed{4} + 1 \quad \boxed{3} + 3 \quad \boxed{1} &= 14 <= 15 \end{aligned}$$

Ejemplo ilustrativo:

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3 \quad \max \quad Z = 3 \boxed{5} + 5 \boxed{2} + 2 \boxed{1} = 27$$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Sujeto a:

$$\begin{aligned} 1 \quad \boxed{5} + 2 \quad \boxed{2} + 1 \quad \boxed{1} &= 10 <= 10 \\ 2 \quad \boxed{5} + 1 \quad \boxed{2} + 3 \quad \boxed{1} &= 15 <= 15 \end{aligned}$$

Ejemplo ilustrativo:

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3 \quad \max \quad Z = 3 \boxed{6.667} + 5 \boxed{1.667} + 2 \boxed{0} = 28.33$$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Sujeto a:

$$\begin{aligned} 1 \quad \boxed{6.667} + 2 \quad \boxed{1.667} + 1 \quad \boxed{0} &= 10 <= 10 \\ 2 \quad \boxed{6.667} + 1 \quad \boxed{1.667} + 3 \quad \boxed{0} &= 15 <= 15 \end{aligned}$$

Probamos con $x_1=1$ $x_2=1$ $x_3=1$
se cumplen holgadamente las
restricciones con $Z=10$

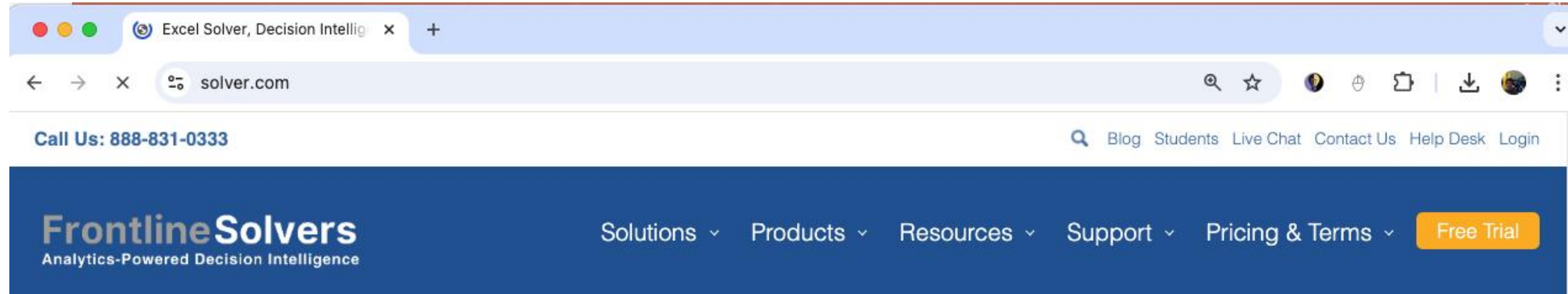
Probamos con $x_1=4$ $x_2=3$ $x_3=1$
no se cumple la restricción 1 por 1
se cumple la restricción 2 por -1
con $Z=29$

Probamos con $x_1=5$ $x_2=2$ $x_3=1$
se cumple la restricción 1 por 0
se cumple la restricción 2 por 0
con $Z=27$

Probamos con $x_1=6.66$ $x_2=1.66$ $x_3=0$
se cumple la restricción 1 por 0
se cumple la restricción 2 por 0
con $Z=28.33$

¿será el óptimo?

¿cómo resolvemos el problema directamente sin tantear (ensayo-y-error)?



The Decision Intelligence Tool You Can Use for Real Results

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- ✓ Do It Yourself in Python, C#, C++, *JavaScript*
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- ✓ Data Science and Machine Learning
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Unlock **all three** capabilities in one: Optimization, Simulation, **AND** Data Science, **plus** DMN Business Rules

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Save 20%

Excel Solver Parameters dialog box is open, showing the following settings:

- Set Objective:** 'LP-primal'!\$Y\$9
- To:** Max
- By Changing Variable Cells:** \$Q\$9:\$T\$9,\$W\$9
- Subject to the Constraints:** \$Y\$14:\$Y\$15 <= \$AA\$14:\$AA\$15
- ☒ Make Unconstrained Variables Non-Negative
- Select a Solving Method:** Simplex LP

The spreadsheet content shows the following linear programming problem:

Ejemplo ilustrativo:

Maximizar $Z = 3x_1 + 5x_2 + 2x_3$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

The optimal solution values are displayed in the spreadsheet:

$$Z = 3(6.667) + 5(1.667) + 2(0) = 28.33$$

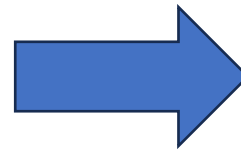
The constraints are also shown with their respective values:

$$\begin{aligned} 1(6.667) + 2(1.667) + 1(0) &= 10 \leq 10 \\ 2(6.667) + 1(1.667) + 3(0) &= 15 \leq 15 \end{aligned}$$

Solver de Excel para obtener los valores óptimos de x_1 , x_2 y x_3

PRIMAL

$$\begin{aligned} &\text{minimizar} && \mathbf{c}^T \mathbf{x} \\ &\text{sujeto a} && \mathbf{Ax} \leq \mathbf{b} \\ &&& \mathbf{x} \geq \mathbf{0} \end{aligned}$$



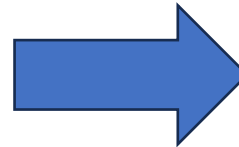
DUAL

$$\begin{aligned} &\text{maximizar} && \lambda^T \mathbf{b} \\ &\text{sujeto a} && \lambda^T \mathbf{A} \leq \mathbf{c}^T \\ &&& \lambda \leq \mathbf{0} \end{aligned}$$

$$\text{Maximizar } Z = 3x_1 + 5x_2 + 2x_3$$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$



$$\text{Minimizar } W = 10y_1 + 15y_2$$

Sujeto a:

$$\begin{aligned} y_1 + 2y_2 &\geq 3, \\ 2y_1 + y_2 &\geq 5, \\ y_1 + 3y_2 &\geq 2, \\ y_1, y_2 &\geq 0. \end{aligned}$$

Excel interface showing a linear programming problem and its solution.

Maximizar $Z = 3x_1 + 5x_2 + 2x_3$ $Z = 3(6.667) + 5(1.667) + 2(0) = 28.33$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Sujeto a:

$$\begin{aligned} 1(6.667) + 2(1.667) + 1(0) &= 10 \leq 10 \\ 2(6.667) + 1(1.667) + 3(0) &= 15 \leq 15 \end{aligned}$$

Solver Results

Solver found a solution. All constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

Reports

☐ Answer
☒ Sensitivity
☐ Limits

☐ Return to Solver Parameters Dialog
☐ Outline Reports

Save Scenario... Cancel OK

Para obtener los precios sombra emitir el reporte SENSITIVITY

Ejemplo ilustrativo:

Sujeto a:

Maximizar $Z = 3x_1 + 5x_2 + 2x_3$ max $Z = 3 \cdot 6.667 + 5 \cdot 1.667 + 2 \cdot 0 = 28.33$

Sujeto a:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 10, \\ 2x_1 + x_2 + 3x_3 &\leq 15, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

$$\begin{aligned} 1 \cdot 6.667 + 2 \cdot 1.667 + 1 \cdot 0 &= 10 \leq 10 \\ 2 \cdot 6.667 + 1 \cdot 1.667 + 3 \cdot 0 &= 15 \leq 15 \end{aligned}$$

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.78 Sensitivity Report							
2	Worksheet: [M1_2_PL_E1.xlsx]LP-primal							
3	Report Created: 4/4/25 10:11:51 AM							
4								
5								
6	Variable Cells							
7								
8	Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease	
9	\$Q\$9	=	6.666666667	0	3	7	0.5	
10	\$T\$9	+	1.666666667	0	5	1	3.5	
11	\$W\$9	+	0	-1.333333333	2	1.333333333	1E+30	
12								
13	Constraints							
14								
15	Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease	
16	\$Y\$14	=	10	2.333333333	10	20	2.5	
17	\$Y\$15	=	15	0.333333333	15	5	10	
18								

PRECIOS SOMBRA
 $\lambda_1 = 2.333$
 $\lambda_2 = 0.333$

El planteamiento dual es:

$$\text{Minimizar } W = 10y_1 + 15y_2 \quad (8)$$

Sujeto a:

$$y_1 + 2y_2 \geq 3,$$

$$2y_1 + y_2 \geq 5,$$

$$y_1 + 3y_2 \geq 2,$$

$$y_1, y_2 \geq 0.$$

Solución dual es $(y_1, y_2) = (2.333, 0.333)$ con $W = 28.333$

Resultados en: https://github.com/pmdeoliveiradejesus/AGyCE/tree/main/M1_Optimization

El planteamiento dual es:

Minimizar $W = 10y_1 + 15y_2$

min $W = 10 \cdot 2.333 + 15 \cdot 0.333 = 28.33$

Sujeto a:

Sujeto a:

$$y_1 + 2y_2 \geq 3,$$

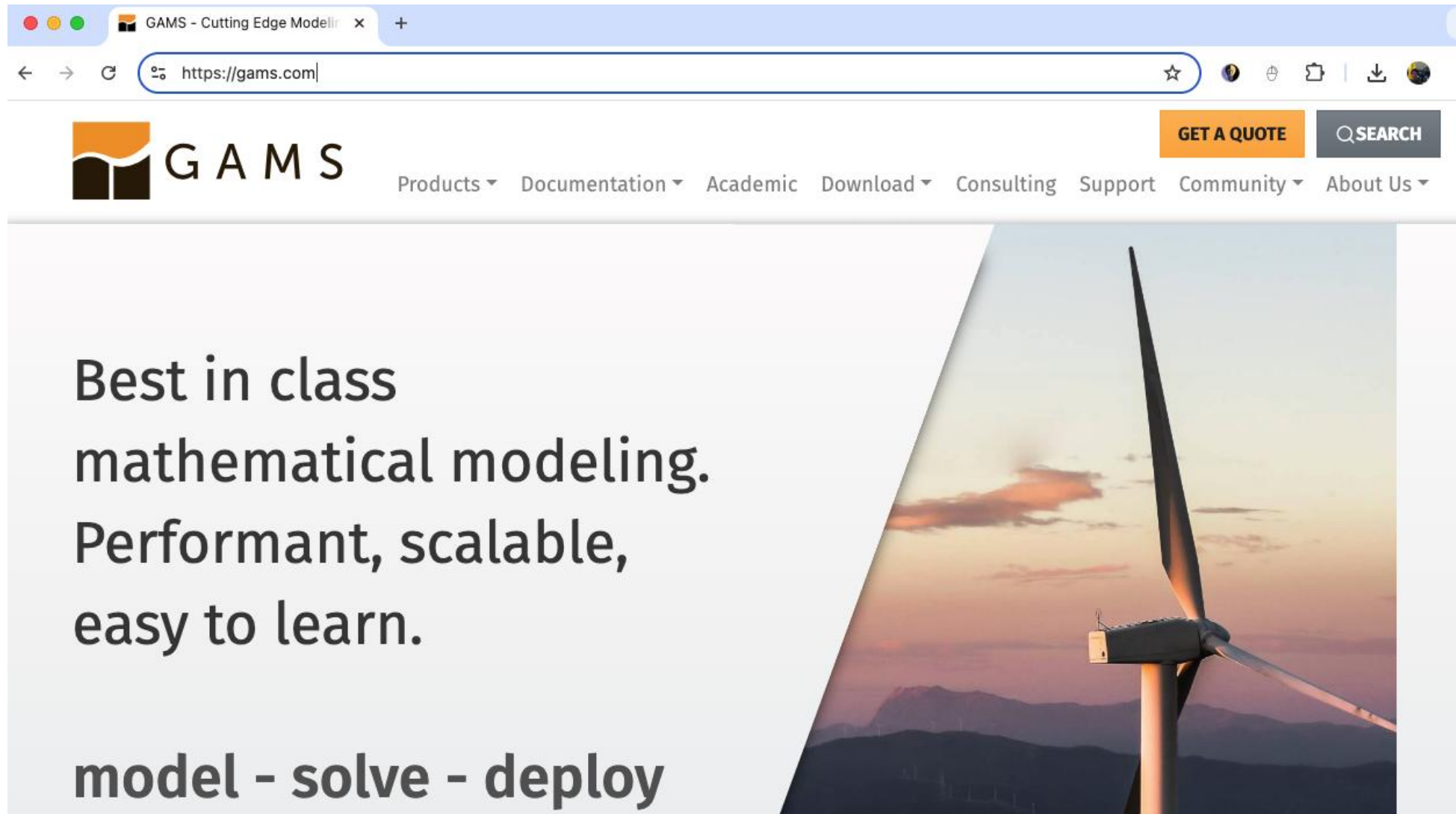
$$2y_1 + y_2 \geq 5,$$

$$y_1 + 3y_2 \geq 2,$$

$$y_1, y_2 \geq 0.$$

$$\begin{array}{rclclcl} 1 & 2.333 & + & 2 \cdot 0.333 & = & 3 & \geq & 3 \\ 2 & 2.333 & + & 1 \cdot 0.333 & = & 5 & \geq & 5 \\ 1 & 2.333 & + & 3 \cdot 0.333 & = & 3.333 & \geq & 2 \end{array}$$

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.78 Sensitivity Report							
2	Worksheet: [M1_2_PL_E1.xlsx]LP-dual							
3	Report Created: 4/4/25 10:20:34 AM							
4								
5								
6	Variable Cells							
7								
8	Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease	
9	\$Q\$9	=	2.333333333	0	10	20	2.5	
10	\$T\$9	+	0.333333333	0	15	5	10	
11								
12	Constraints							
13								
14	Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease	
15	\$V\$14	=	3	6.666666667	3	7	0.5	
16	\$V\$15	=	5	1.666666667	5	1	3.5	
17	\$V\$16	=	3.333333333	0	2	1.333333333	1E+30	
18								



The image is a screenshot of a web browser displaying the GAMS website. The browser's address bar shows the URL <https://gams.com>. The website's header features the GAMS logo on the left, a navigation menu with links for Products, Documentation, Academic, Download, Consulting, Support, Community, and About Us in the center, and two buttons labeled 'GET A QUOTE' and 'SEARCH' on the right. The main content area has a light blue background with a large, stylized image of a wind turbine on the right side. Overlaid on the left side of this image is the text: 'Best in class mathematical modeling. Performant, scalable, easy to learn.' and 'model - solve - deploy'.

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	License ▼	LP	MIP	RMIP	NLP	MCP	MPEC	RMPEC	CNS	DNLP	RMI ▲
ALPHAECIP	Demo										
ANTIGONE	Demo				X				X	X	X
BARON	Demo	X	X	X	X				X	X	X
CBC	Demo	X	X	X							
CONOPT	Demo	X		X	X				X	X	X
CONOPT4	Demo	X		X	X				X	X	X
CONVERT	Demo	X	X	X	X	X	X	X	X	X	X
COPT	Demo	X	X	X							
CPLEX	Demo	X	X	X							
DE	Demo										
DECISC	Demo	X									
DECISM	Demo	X									
DICOPT	Demo										
EXAMINER	Demo	X	X	X	X	X	X	X		X	X
GAMSCHK	Demo	X	X	X	X	X				X	X
GUROBI	Demo	X	X	X	X					X	X
HIGHS	Demo	X	X	X							
IPOPT	Demo	X		X	X				X	X	X

https://en.wikipedia.org/wiki/General_algebraic_modeling_system

GAMS/Solver Prices & Price per User				
Type of License	Single user	5-users / \$pp	10-users / \$pp	+20-users / \$pp
Academic ⁸ – “free” solvers ⁴	\$640	\$256	\$192	\$128
Academic ⁸ – CPLEX-Link ^{6,7}	\$1,280	\$512	\$384	\$256
Academic ⁸ – full CPLEX	\$1,920	\$768	\$576	\$384
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Non-Academic – “free” solvers ⁴ Appl ⁵	\$1,600	\$640	\$480	\$320
Non-Academic – CPLEX-Link ⁷	\$6,400	\$2,560	\$1,920	\$1,280
Non-Academic – Full CPLEX Appl ⁵	\$6,400	\$2,560	\$1,920	\$1,280
Non-Academic – CPLEX-Link ⁷ Appl ⁵	\$4,800	\$1,920	\$1,440	\$960
Non-Academic – Full CPLEX or XPRESS	\$12,800	\$5,120	\$3,840	\$2,560

Ejemplo 1 – solución en GAMS

```
Welcome | M1_1_E1.gms | Results M1_1_LP_E1.gdx
1 $title (LP model)
2 * Paulo M. De Oliveira-De Jesus pdeoliv@gmail.com
3 Variable x1, x2, x3, z;
4 Positive Variables x1, x2, x3;
5 Equation eq1, eq2, eq3;
6 eq1.. 1 * x1 + 2 * x2 + 1 * x3 =l= 10;
7 eq2.. 2 * x1 + 1 * x2 + 3 * x3 =l= 15;
8 eq3.. 3 * x1 + 5 * x2 + 2 * x3 =e= z;
9 Model LP / all /;
10 solve LP using lp maximizing z;
11 display x1.l, x2.l, x3.l, z.l;
12 execute_unload 'Results M1_1_LP_E1.gdx';
```

GAMS 51.2.1 5fb43f63 Oct 13, 2025 DEX-DEG x86 64bit/macOS
(LP model)
Execution

----	11	VARIABLE x1.L	=	6.667
		VARIABLE x2.L	=	1.667
		VARIABLE x3.L	=	0.000
		VARIABLE z.L	=	28.333

EXECUTION TIME = 0.749 SECONDS 3 MB 51.2.1 5fb43

Archivo LST

Entry	Name ^	Type
5	eq1	Equation (=l=)
6	eq2	Equation (=l=)
7	eq3	Equation (=e=)
1	x1	Variable (Positive)
2	x2	Variable (Positive)
3	x3	Variable (Positive)
4	z	Variable (Free)

Level	Marginal
10	2.33333

Entry	Name ^	Type
5	eq1	Equation (=l=)
6	eq2	Equation (=l=)
7	eq3	Equation (=e=)
1	x1	Variable (Positive)
2	x2	Variable (Positive)
3	x3	Variable (Positive)
4	z	Variable (Free)

Level	Marginal
15	0.333333

Archivo GDX

En Python (Gurobi, propietario)

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Ejemplo 1 – Solución en Python (SciPy, gratis)

Gemini prompt: Give me a python script to solve the primal and dual for the following LP problem: max $3x_1 + 5x_2 + 2x_3$, s.t. $x_1 + 2x_2 + x_3 \leq 10$; $2x_1 + x_2 + 3x_3 \leq 15$; x_1, x_2, x_3 are positive

```
import numpy as np
from scipy.optimize import linprog

# -----
# 1. Solve the Primal Problem
# -----
# max  $3x_1 + 5x_2 + 2x_3$ 
# s.t.
#  $x_1 + 2x_2 + x_3 \leq 10$ 
#  $2x_1 + x_2 + 3x_3 \leq 15$ 
#  $x_1, x_2, x_3 \geq 0$ 

print("--- 1. Solving the Primal Problem ---")

# Objective function coefficients
```

Maximizar $Z = 3x_1 + 5x_2 + 2x_3$

Sujeto a:

$$x_1 + 2x_2 + x_3 \leq 10,$$

$$2x_1 + x_2 + 3x_3 \leq 15,$$

$$x_1, x_2, x_3 \geq 0.$$

can you give me a Jupyter Notebook script to solve this problem y Google Colab?

3. Programación Cuadrática

El objetivo de la programación cuadrática es encontrar un vector x de dimensión n que minimice la función de costo:

$$f(x) = \frac{1}{2}x^T Q x + c^T x$$

sujeto a restricciones lineales y/o no-lineales cuadráticas:

$$Ax \preceq b$$

$$\frac{1}{2}x^T Q_r x + d^T x$$

donde x^T denota la transposición vectorial de x , y $\preceq$ representa la desigualdad por componentes (es decir, cada entrada del vector (Ax) es menor o igual que la entrada correspondiente del vector b . d es un vector real de dimensión n . Q_r es una matriz simétrica real de dimensión $(n \times n)$.

Ejemplo 2 - NLP/QP

Ejemplo ilustrativo de la solución de un problema NLP-QP:

$$\text{Minimizar } Z = 2x_1^2 + x_1x_2 + x_2^2 + \frac{3}{2}x_3^2 + x_1 - 2x_2$$

$$Z = \frac{1}{2} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^T \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Sujeto a:

$$\begin{aligned} x_1^2 + x_2^2 + x_3^2 &= 4 \\ (x_1 - 1)^2 + x_2^2 &\leq 1 \end{aligned}$$

Las ecuaciones 16-21 constituyen un sistema de 5 ecuaciones no-lineales con 5 incógnitas.
La solución óptima es:

$$x_1=0.344, x_2=0.754, x_3=1.820, \lambda=-1.500, \psi=0.000, \mu=1.598 \text{ con } Z = 4.868.$$

Resultados en: https://github.com/pmdeoliveiradejesus/AGyCE/tree/main/M1_Optimization

Problema de Despacho Económico (DE)

¿cómo producir electricidad de forma eficiente?

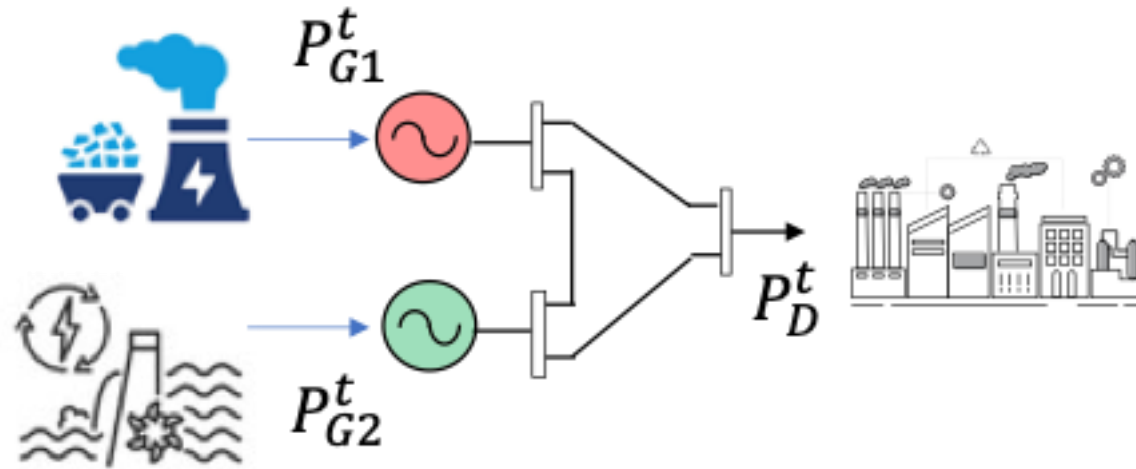
El Despacho Económico (DE) permite asignar los bloques de generación

A mínimo costo de producción (combustible)

Condicionado a:

Satisfacer la demanda de forma **SEGURA** en régimen estacionario

Despacho Económico Centralizado (DE)



$$\text{minimize} \quad \sum_{t=1}^{24} \sum_{k=1}^2 C(P_{Gk}^t)$$

s.t.

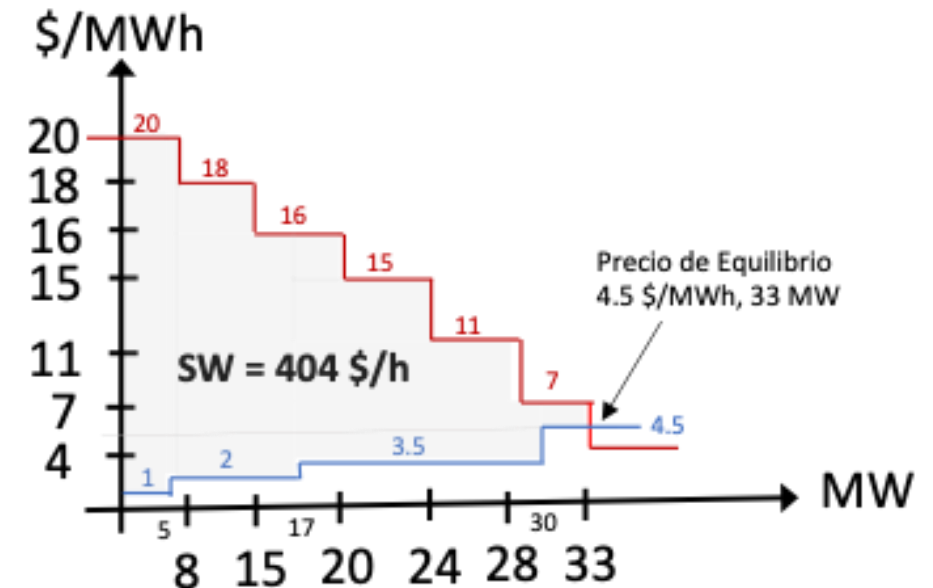
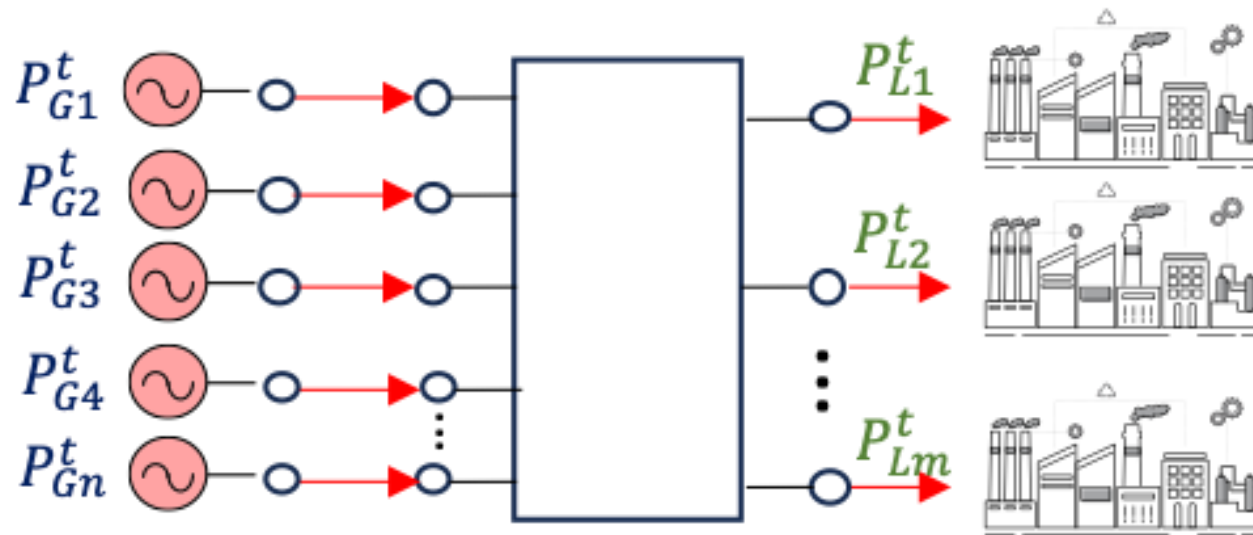
$$\sum_{k=1}^2 P_{Gk}^t = P_D^t \quad \forall t = 1, \dots, 24$$

PRECIO MARGINAL HORARIO DEL SISTEMA

Solución dual

(Multiplicador de Lagrange o Precio Sombra)
asociado a la ecuación de balance de potencia

Despacho Económico mediante Subastas Bolsa de Energía



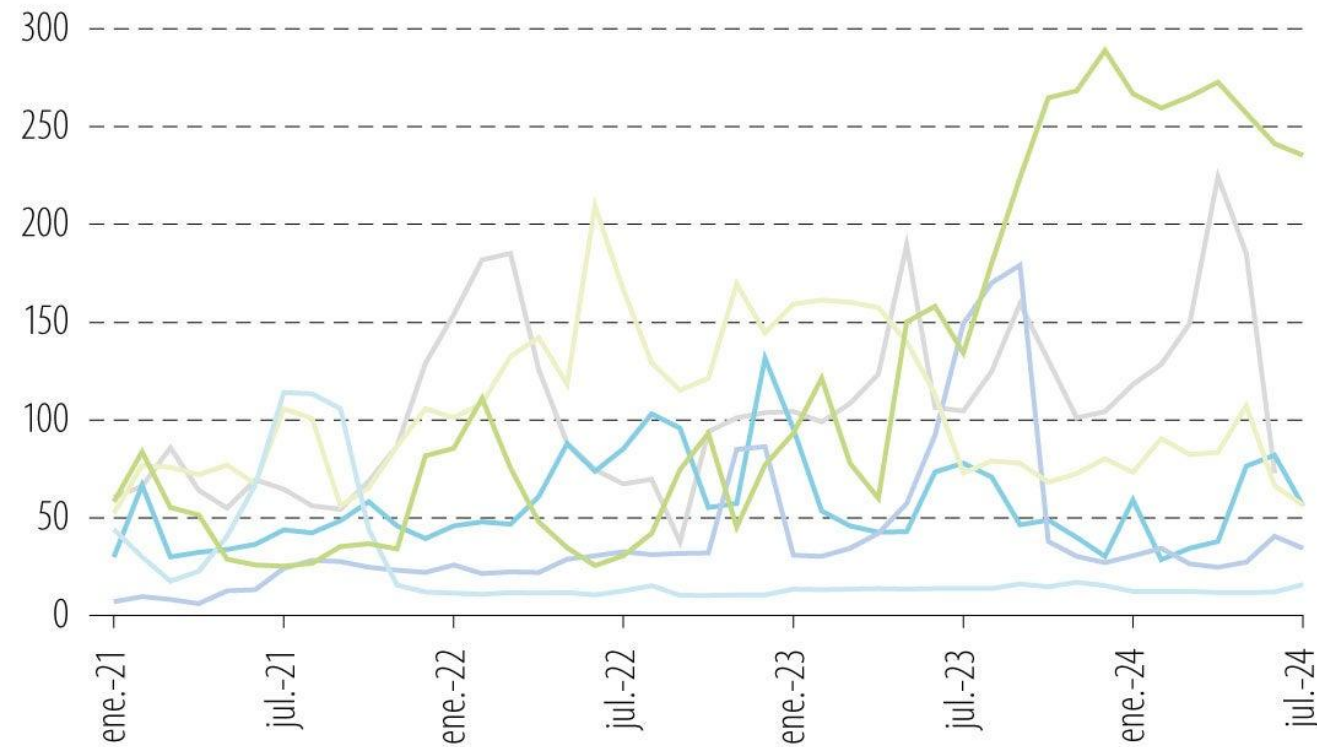
Despacho Económico en Colombia - XM

<https://www.youtube.com/watch?v=uK-hnIU2ReA>

PRECIOS SPOT DE ENERGÍA EN AMÉRICA LATINA

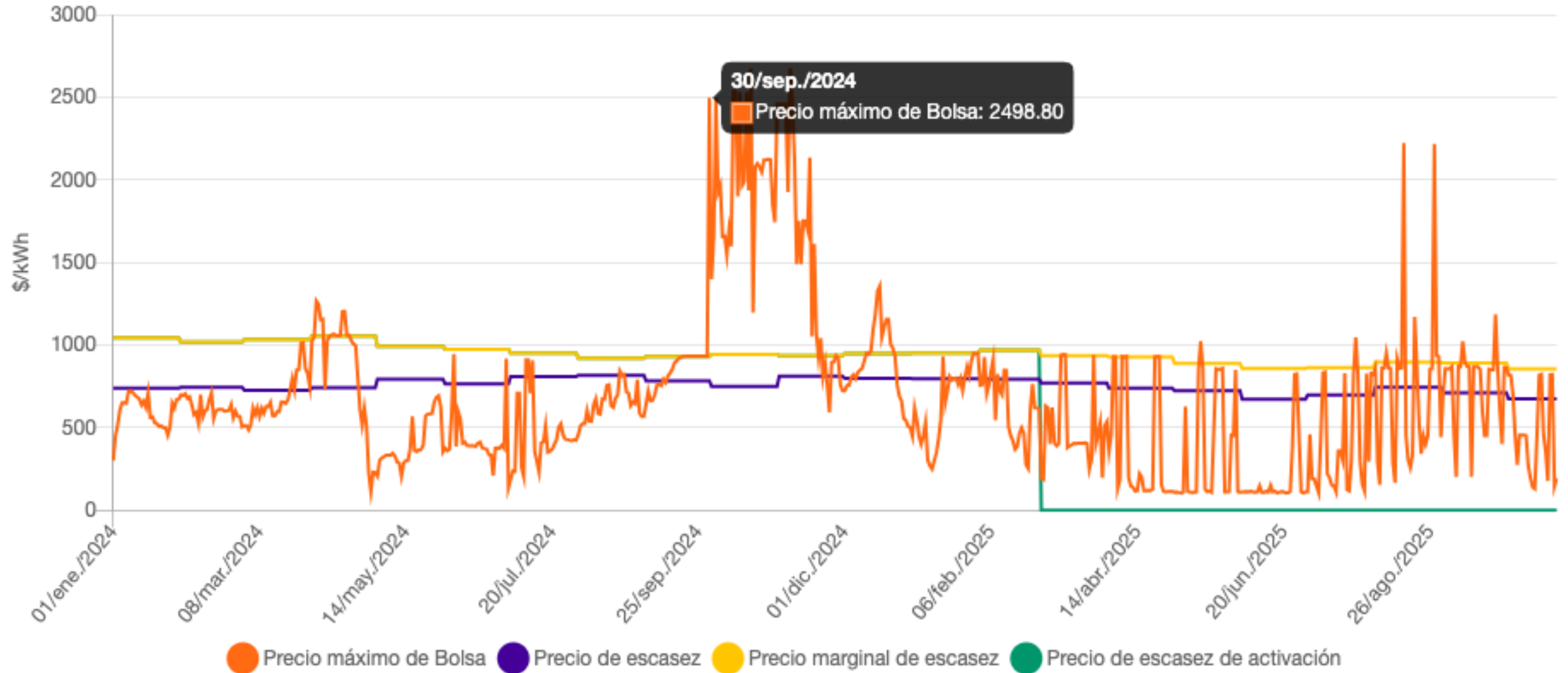
— Panamá — Perú — Brasil — México — Chile — Colombia

(US\$/MWh)

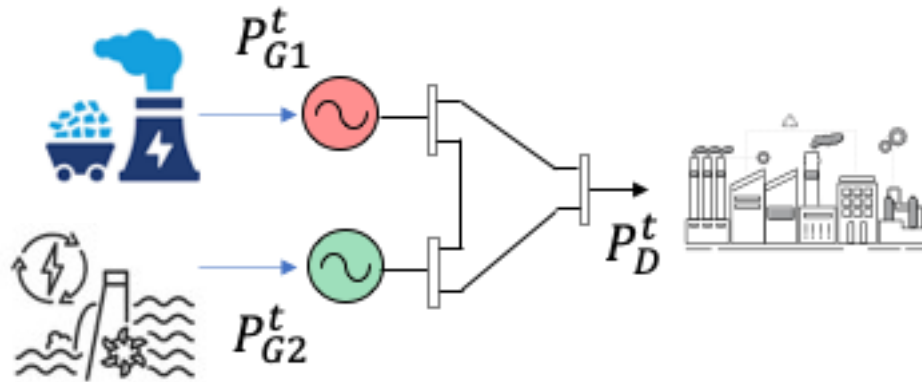


Fuente: S&P Global Ratings / Gráfico: LR-ER



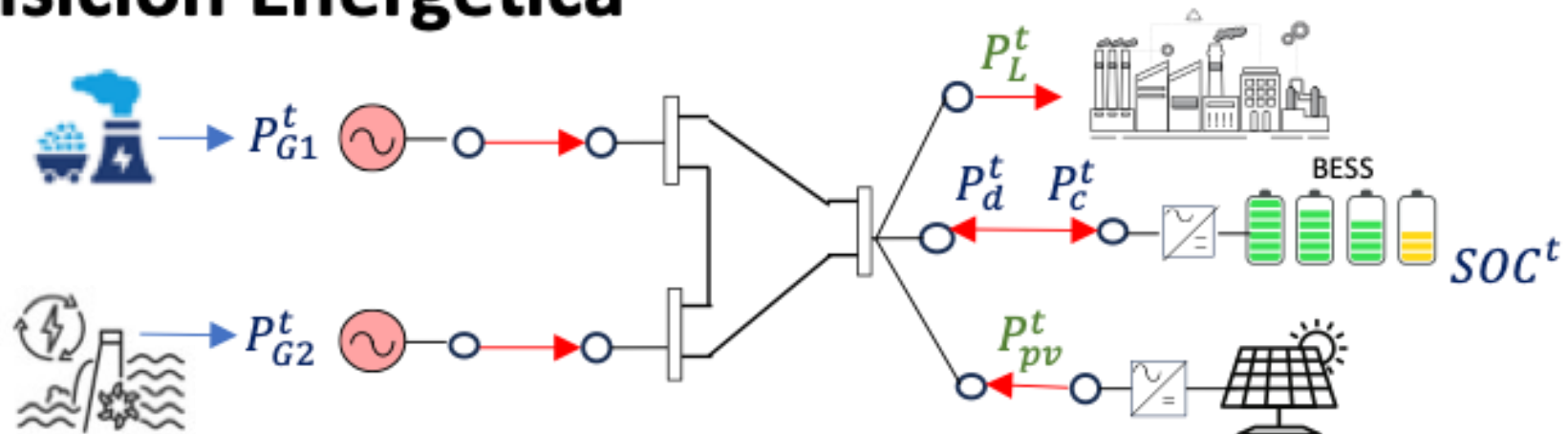


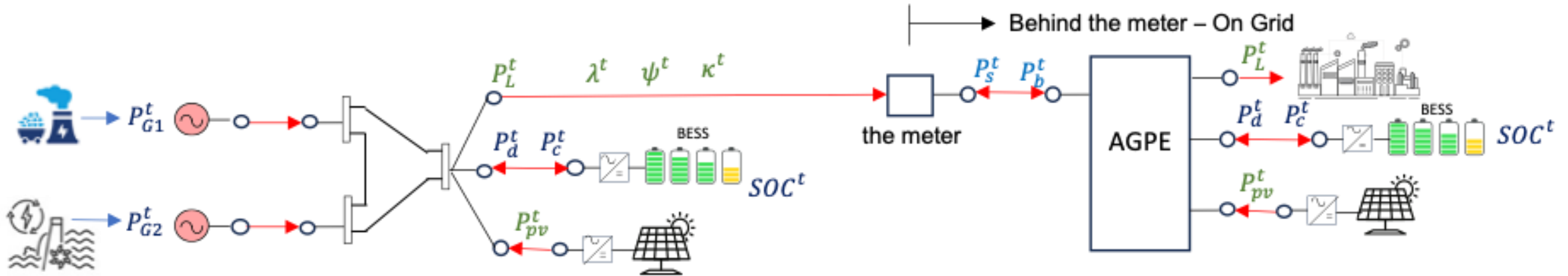
Despacho Económico Hidro-Térmico (HTD)



$$\begin{aligned} \min \quad & \sum_{t=1}^{24} \sum_{k=1}^2 C(P_{Gk}^t) \\ \text{s.t.} \quad & \sum_{k=1}^2 P_{Gk}^t = P_D^t \quad \forall t = 1, \dots, 24 \end{aligned}$$

DE en Transición Energética





DE Centralizado en Transición Energética

Minimizar Costo Social
Sujeto a
Leyes Físicas (KCL, KVL, 2da Ley Termodinámica)
Seguridad del Sistema

DE Descentralizado en Transición Energética

Maximizar Beneficio Económico del AGPE
Sujeto a
Leyes Físicas (KCL, KVL, 2da Ley Termodinámica)

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10857878>

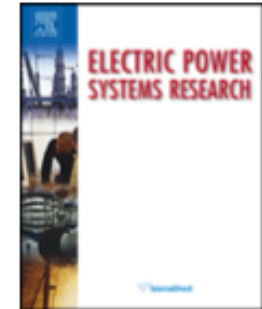
The Rooftop Revolution – António Gómez Expósito (2025)



Contents lists available at ScienceDirect

Electric Power Systems Research

journal homepage: www.elsevier.com/locate/epsr



Optimal energy exchange of an industrial cogeneration in a day-ahead electricity market

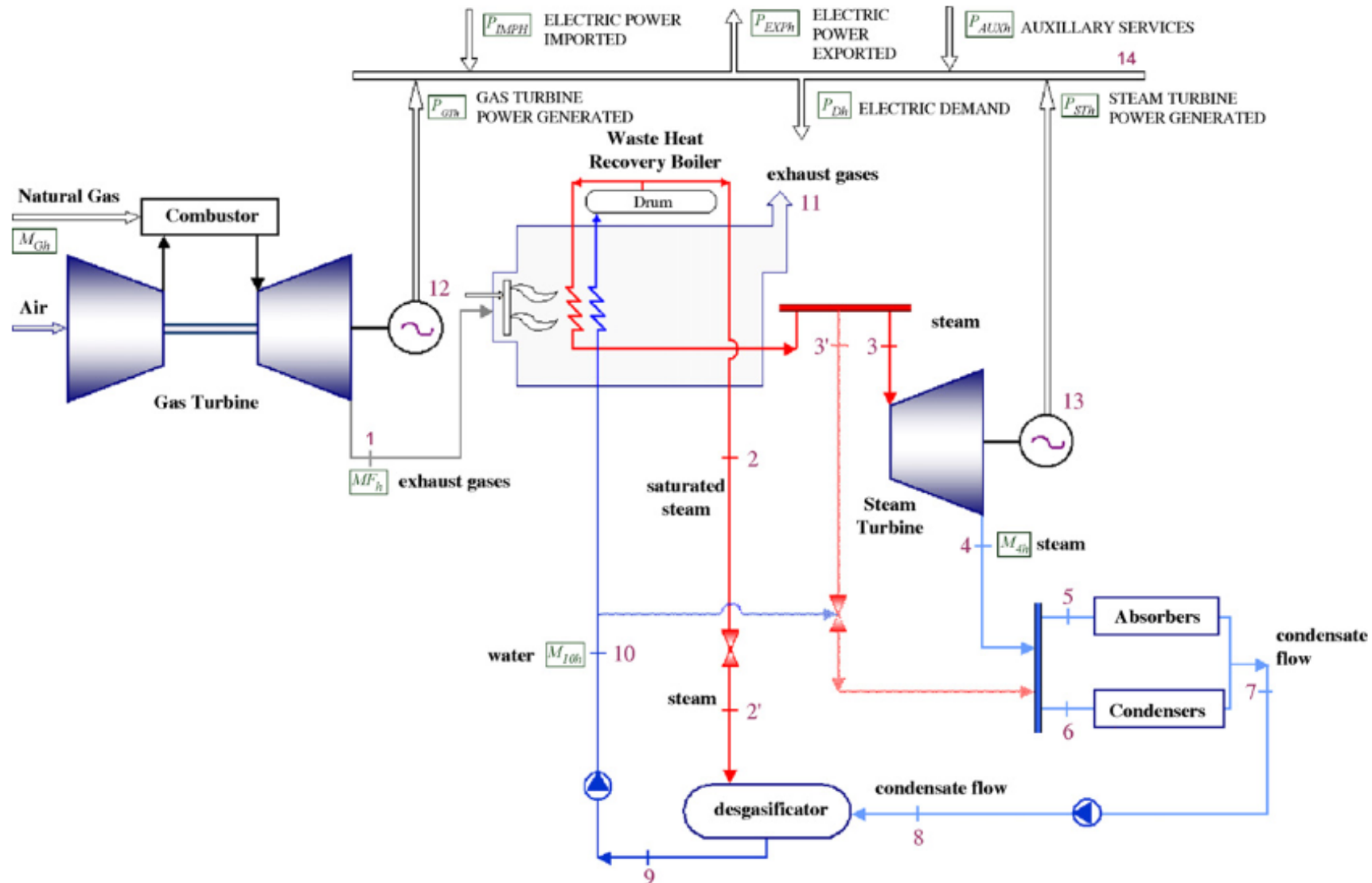
J.M. Yusta^{a,*}, P.M. De Oliveira-De Jesus^b, H.M. Khodr^c

^a Department of Electrical Engineering, University of Zaragoza, Spain

^b Department of Electrical and Computer Engineering of Faculdade de Engenharia da Universidade do Porto, Portugal

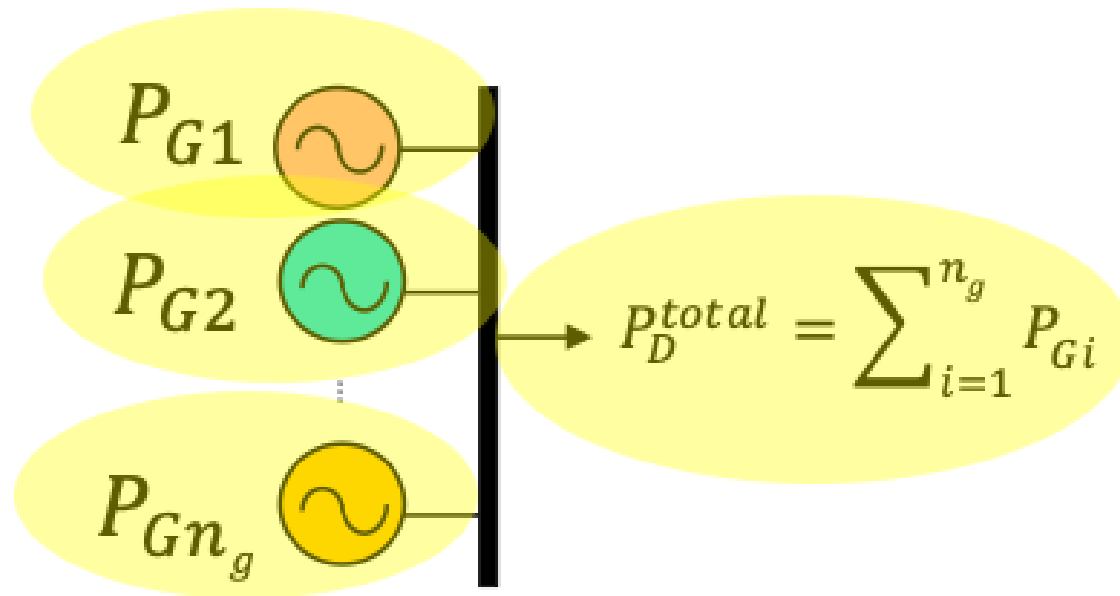
^c Knowledge Engineering and Decision-Support Research Group of the Electrical Engineering Institute of Porto, Portugal

<https://github.com/pmdeoliveiradejesus/AGyCE/blob/main/M2%20Despacho%20Econ%C3%B3mico/CogenBtM.pdf>



Problema de Despacho Económico (DE)

Supongamos un sistema de n_g unidades generadoras alimenta una demanda



$$\min_{P_G} C(P_G) = \sum_{i=1}^{n_g} C_i(P_{Gi})$$

$$\sum_{i=1}^{n_g} P_{Gi} = P_D^{total}$$

El despacho económico consiste en hallar el arreglo de producción P_G y el precio λ que permita minimizar los costos operacionales totales.

Formulación del problema

El problema de optimización económica consiste encontrar el despacho de potencia activa en n_g generadores $\mathbf{P}_G = [P_{G1}, \dots, P_{Gn_g}]^T$ tal que satisfaga la demanda P_D^{total} a mínimo costo de producción:

$$\min_{\mathbf{P}_G} C(\mathbf{P}_G) = \sum_{i=1}^{n_g} C_i(P_{Gi}) \quad (1)$$

sujeto a:

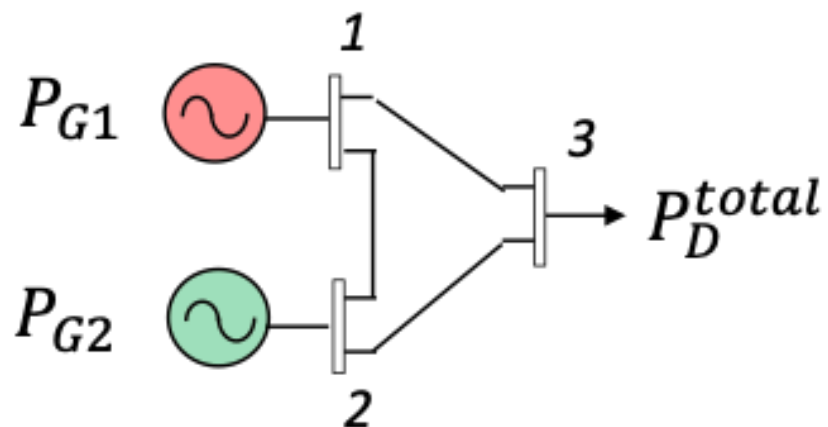
$$\sum_{i=1}^{n_g} P_{Gi} = P_D^{\text{total}} \quad (2)$$

La función de costo de producción de cada generador i es:

$$C_i(P_{Gi}) = C_{0i} + a_i P_{Gi} + \frac{1}{2} b_i P_{Gi}^2 \quad (3)$$

donde C_0 , a y b son coeficientes de la función de costo dada en la Ec. 3. Los parámetros dependen de la tecnología de generación utilizada. Para la interpretación económica de los factores C_0 , a y b revisar sección 1.6 y 3.1 del texto de Wood & Wollemberg (2013) [3].

Ejemplo 3: Problema de Despacho Económico (DE)



Generation data

Unit	Co	a	b
1	100	20	0.05
2	200	25	0.1

Demand data

Pd	600
----	-----

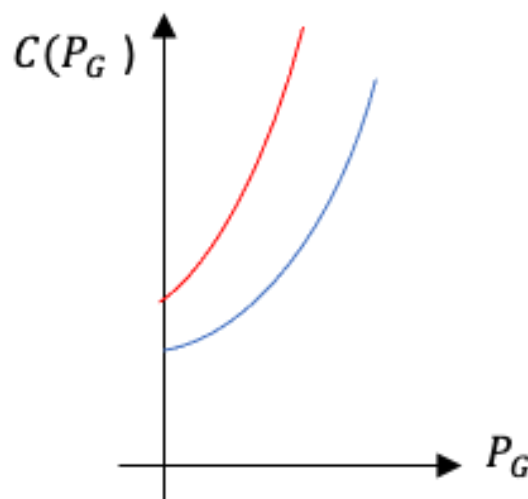
$$\min C(P_{G1}) + C(P_{G2})$$

s.t.

$$P_{G1} + P_{G2} = P_D^{total}$$

$$C(P_{G1}) = C_{01} + a_1 P_{G1} + 0.5b_1 P_{G1}^2$$

$$C(P_{G2}) = C_{02} + a_2 P_{G2} + 0.5b_2 P_{G2}^2$$



**Minimizar costo de producción
sujeto a
Balance de potencia**

PROBLEMA QP

Solución del Problema de Despacho Económico (DE) mediante el Lagrangiano

El Lagrangiano posee la siguiente estructura:

$$\mathcal{L}(P_G, \lambda) = \sum_{i=1}^{n_g} C_i(P_{Gi}) + \lambda \left(P_D^{\text{total}} - \sum_{i=1}^{n_g} P_{Gi} \right)$$

Las condiciones necesarias de primer-orden de optimalidad son:

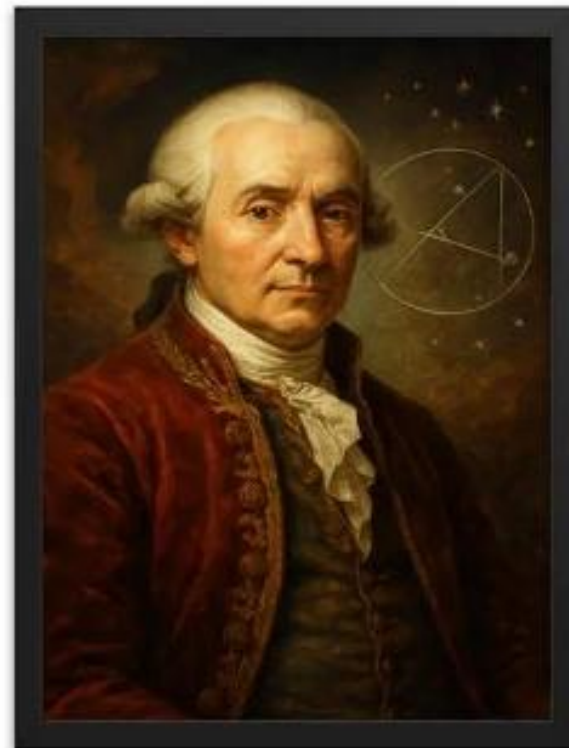
$$\frac{\partial \mathcal{L}(\cdot)}{\partial \lambda} = \sum_{i=1}^n P_{Gi} - P_D^{\text{total}} = 0$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial P_{Gi}} = IC_i(P_{Gi}) - \lambda = 0; \quad i = 1, \dots, n_g$$

donde el costo incremental de cada generador i es

$$IC_i(P_{Gi}) = \frac{dC_i(P_{Gi})}{dP_{Gi}} = a_i + b_i P_{Gi}$$

Las ecuaciones 8 y 9 constituyen un sistema de n_g+1 ecuaciones lineales con n_g+1 incógnitas cuya solución es directa.



Joseph Louis Lagrange

Solución de Gómez Expósito *et al*

En [2] se plantea una elegante solución al problema. Especificando los datos conocidos como vectores:

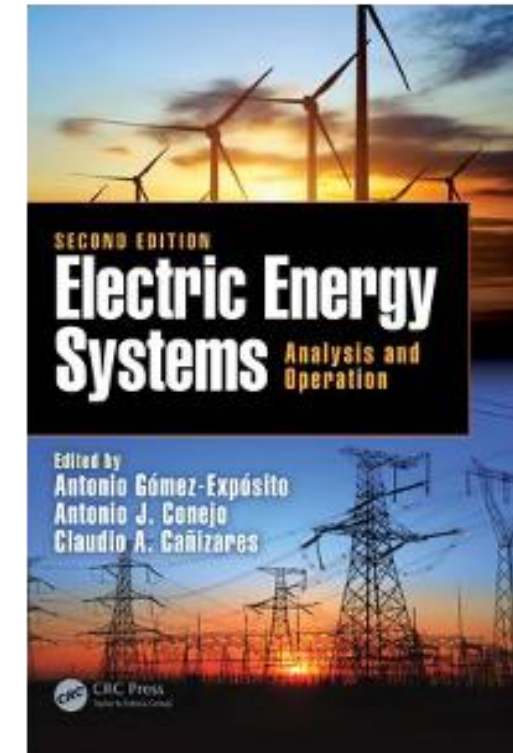
$$\begin{aligned} C_0 &= [C_{01}, \dots, C_{0n_g}]^T \text{ en } \$/\text{h} \\ a &= [a_1, \dots, a_{n_g}]^T \text{ en } \$/\text{MWh} \\ B &= \text{diag}([b_1, \dots, b_{n_g}]^T) \text{ en } \$/\text{MWh}^2 \\ e &= [1, \dots, 1]^T \end{aligned}$$

El costo incremental del sistema y despacho resultante es:

$$\begin{aligned} \lambda &= \frac{P_D^{\text{total}} + e^T B^{-1} a}{e^T B^{-1} e} \text{ en } \$/\text{MWh} \\ P_G &= \frac{B^{-1} e}{e^T B^{-1} e} P_D^{\text{total}} + \frac{B^{-1} e (e^T B^{-1} a)}{e^T B^{-1} e} - B^{-1} a \text{ en MW} \end{aligned}$$

Nótese que el costo fijo C_0 no está incluido en la solución. Esto tiene sentido por cuanto el costo fijo es una constante que desaparece en el análisis incremental.

[2] Antonio Gómez-Expósito, Antonio J Conejo, and Claudio Cañizares, *Electric energy systems: analysis and operation*, CRC press, 2018.



Disponible en

https://uniandes.primo.exlibrisgroup.com/permalink/57U_UDLA/19gnmdh/cdi_askewsholts_vlebooks_9781351754507

Ejemplo 3: Despacho Clásico

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Despacho_Economico_Basico

Home Insert Draw Page Layout Formulas Data Review View Picture Format

Get External Data Refresh All Connections Properties Edit Links Sort Filter Clear Reapply Advanced Text to Columns What-If Analysis Group Ungroup Subtotal Analysis Tools Data Analysis Solver

Picture 5

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Operación Económica de Sistemas de Potencia
Ejemplo 1: Despacho Económico Básico

Database tomada de Caso de Estudio 1

Generation data

Unit	Co \$/h	a \$/MWh	b \$/MWh ²
1	100	20	0.05
2	200	25	0.1

Demand data

Pd 600 MW

Area 1

P_{G1} P_{G2} P_D^{total}

database direct Sensitivity Report 1 NLP +

130%

Consideremos el problema de despacho económico uninodal en el caso de estudio 1¹ La solución de despacho óptimo para la condición de demanda máxima ($P_D^{total}=600$ MW) se muestra en la Tabla 1. Para hallar la solución se aplicó la aproximación compacta (Ec. 17). En este caso las matrices α y β son:

$$\beta = \begin{bmatrix} b_1 & 0 & -1 \\ 0 & b_1 & -1 \\ 1 & 1 & 0 \end{bmatrix} \quad \alpha = [-a_1, -a_2, P_D^{total}]^T$$

La solución es directa:

$$\mathbf{x} = \begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} b_1 & 0 & -1 \\ 0 & b_1 & -1 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} -a_1 \\ -a_2 \\ P_D^{total} \end{bmatrix}$$

Esc.	P_D^{total} MW	P_{G1} MW	P_{G2} MW	λ \$/MWh	C \$/h
max	600	433.3	166.7	41.67	19217
min	250	200	50	30.00	6675

Ejemplo 4: Despacho Clásico con restricciones de capacidad

Despacho_Economico_Capacidad_Generacion

Home Insert Draw Page Layout Formulas Data Review View

Calibri (Body) 12 A A

General

Conditional Formatting Format as Table Cell Styles

Insert Delete Format

Sort & Filter Find & Select

E25

A B C D E F G H I J K L

1 Universidad de los Andes

2 Facultad de Ingeniería

3 Operación Económica de Sistemas de Potencia

4 Ejemplo 2: Despacho Económico Básico con restricciones de capacidad de generación

6

7 Database tomada de Caso de Estudio 1

8

9 Generation data

Unit	Co	a	b	P_{Gmin}	P_{Gmax}
	\$/h	\$/MWh	\$/MWh ²	MW	MW
1	100	20	0.05	0	400
2	200	25	0.1	0	300

16 Demand data

17 Pd 600 MW

18

19

20

database direct Sensitivity Report 1 NLP +

Ready

Area 1

The diagram shows two generators, P_{G1} (red) and P_{G2} (green), connected to a vertical busbar. An arrow points from the busbar to the label P_D^{total} .

Ejemplo 5: Despacho Multinodal con pérdidas (red de transmisión)

Despacho_Economico_Basico_Red

Home Insert Draw Page Layout Formulas Data Review View

Universidad de los Andes
Facultad de Ingeniería

Operación Económica de Sistemas de Potencia
Ejemplo 3: Despacho Económico con Pérdidas (Red de Transmisión)

Database tomada de Caso de Estudio 2

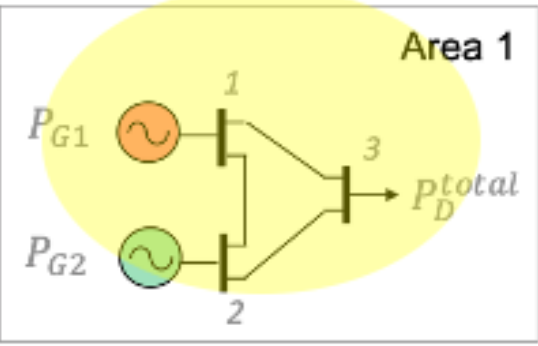
Generation data

Unit	Co	a	b	P_{Gmin}	P_{Gmax}
	\$/h	\$/MWh	\$/MWh ²	MW	MW
1	100	20	0.05	0	400
2	200	25	0.1	0	300

Demand data

P_d 250 MW
 S_{base} 200 MVA

Area 1



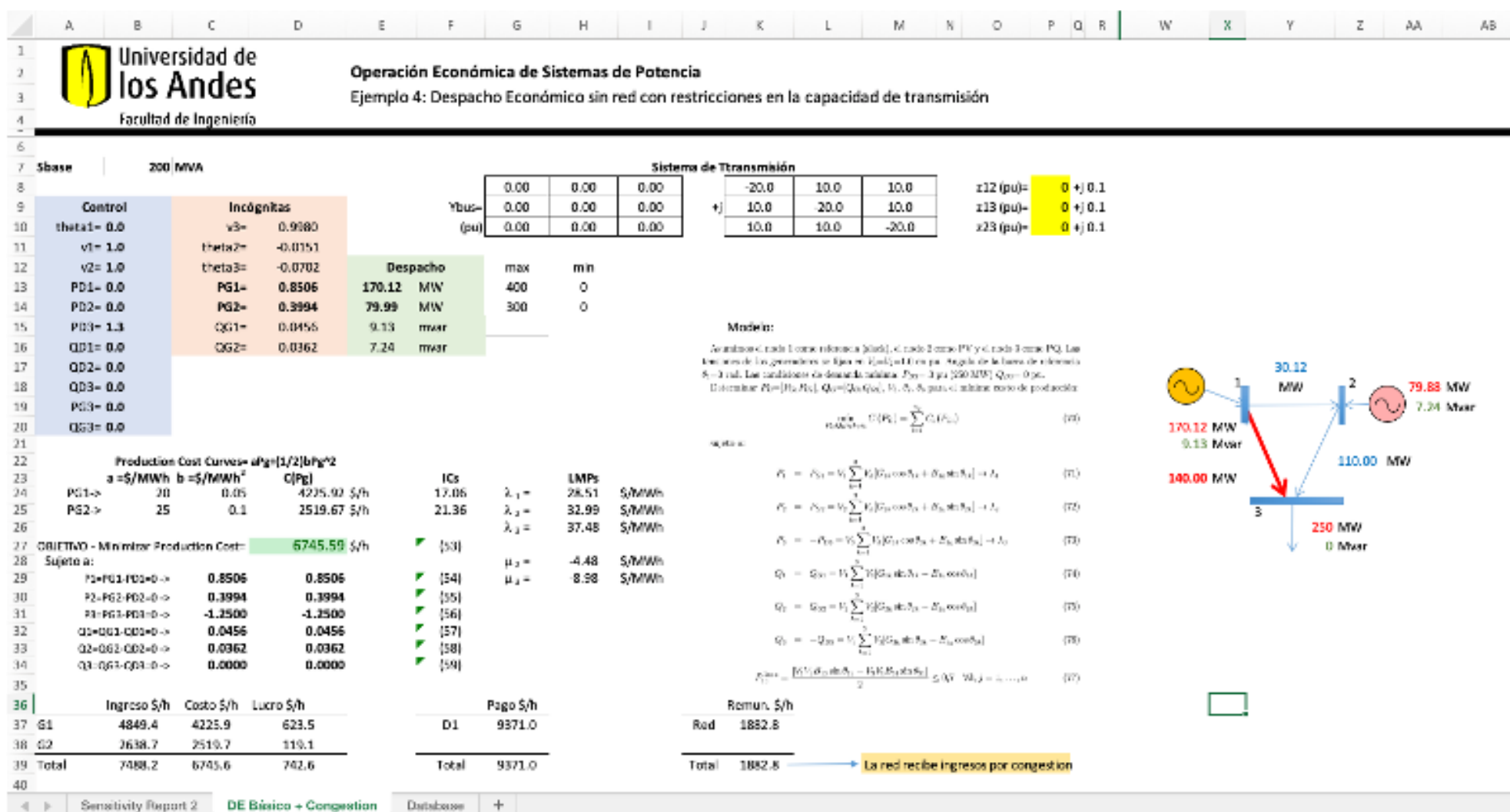
Control

theta1= 0.0
 v1= 1.0
 v2= 1.0
 PD1= 0.0
 PD2= 0.0
 PD3= 250.0
 QD1= 0.0
 QD2= 0.0
 QD3= 0.0
 PG3= 0.0
 QG3= 0.0

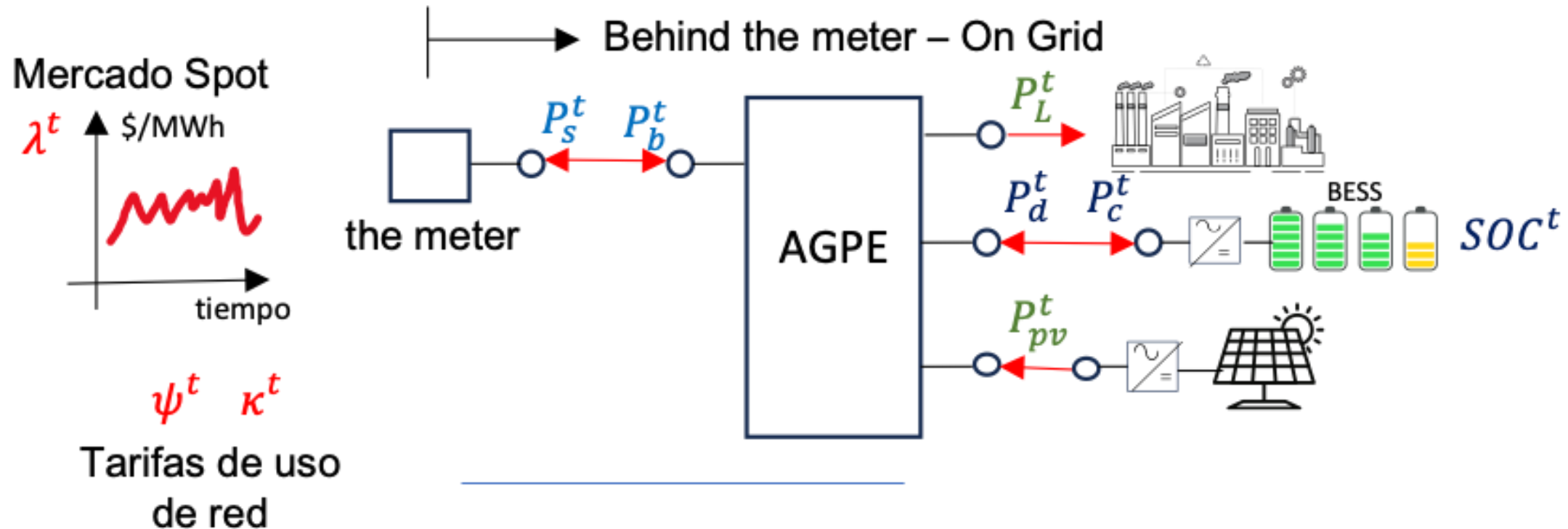
Sensitivity Report 1 DE Básico + Red Database

Ready 150%

Ejemplo 6: Despacho Multinodal sin pérdidas con límites de transmisión

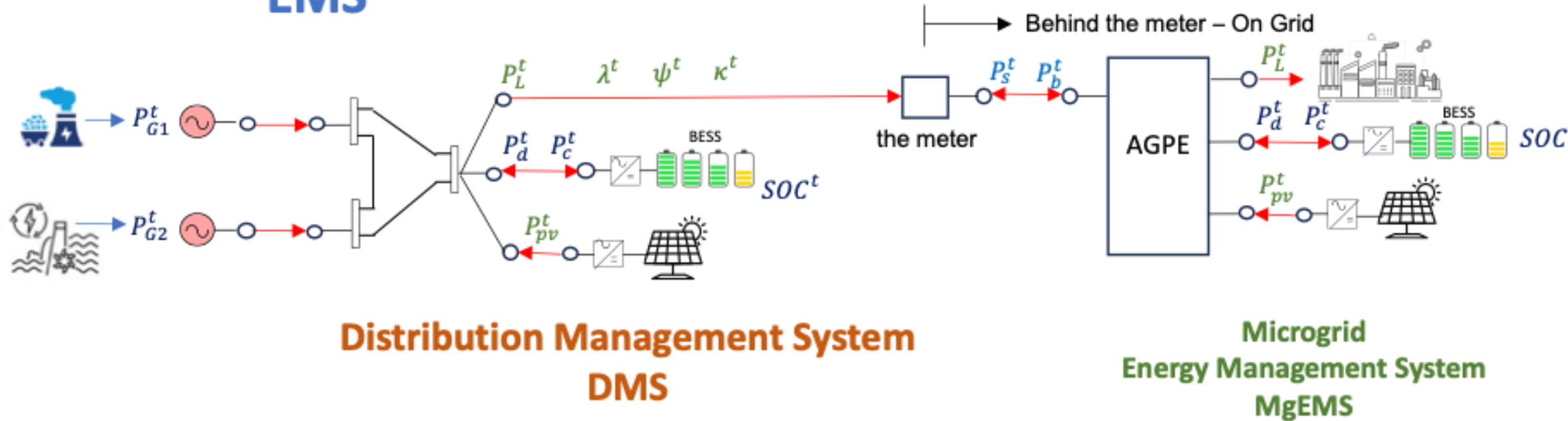


Construyendo modelos de despacho de AGPE

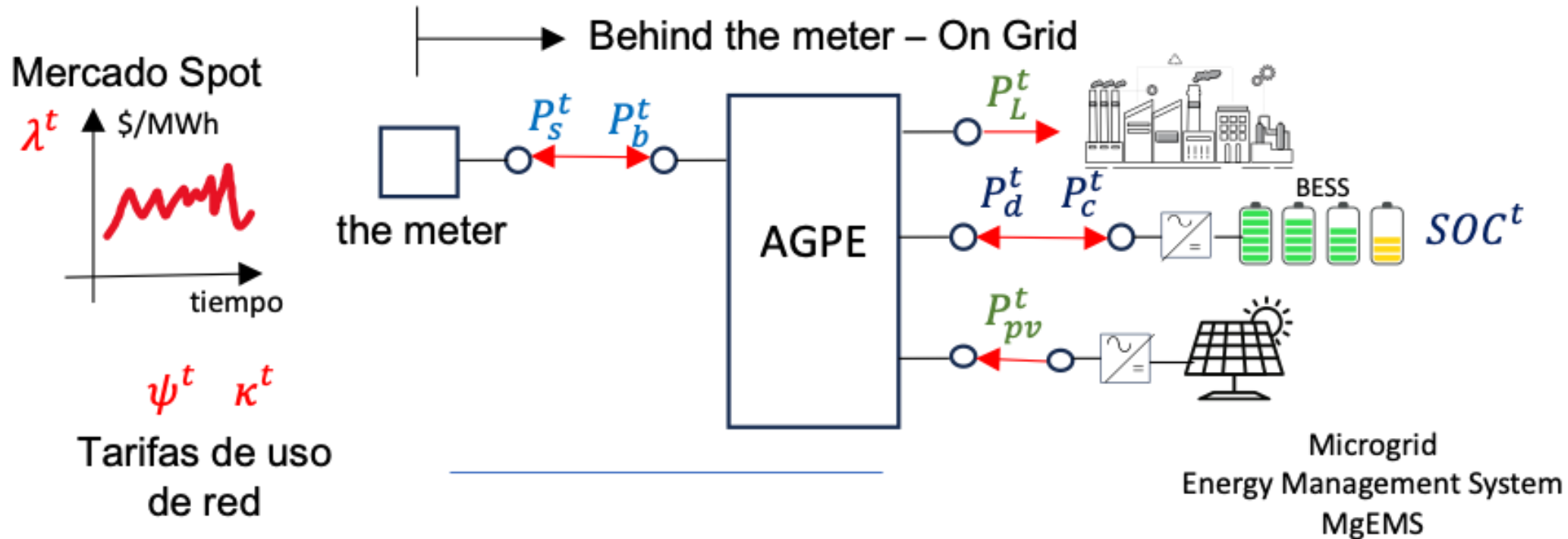


Caracterización de la demanda
 Caracterización del mercado eléctrico
 Requerimientos regulatorios (CREG 174-2021, Colombia)
 Caracterización del recurso solar (pasado, futuro)
 Pre-Dimensionamiento del sistema PV (AGPE < 1000 kW)
 Pre-Dimensionamiento del sistema BESS
Modelo de Despacho óptimo (Forecast 24horas)

Energy Management System EMS

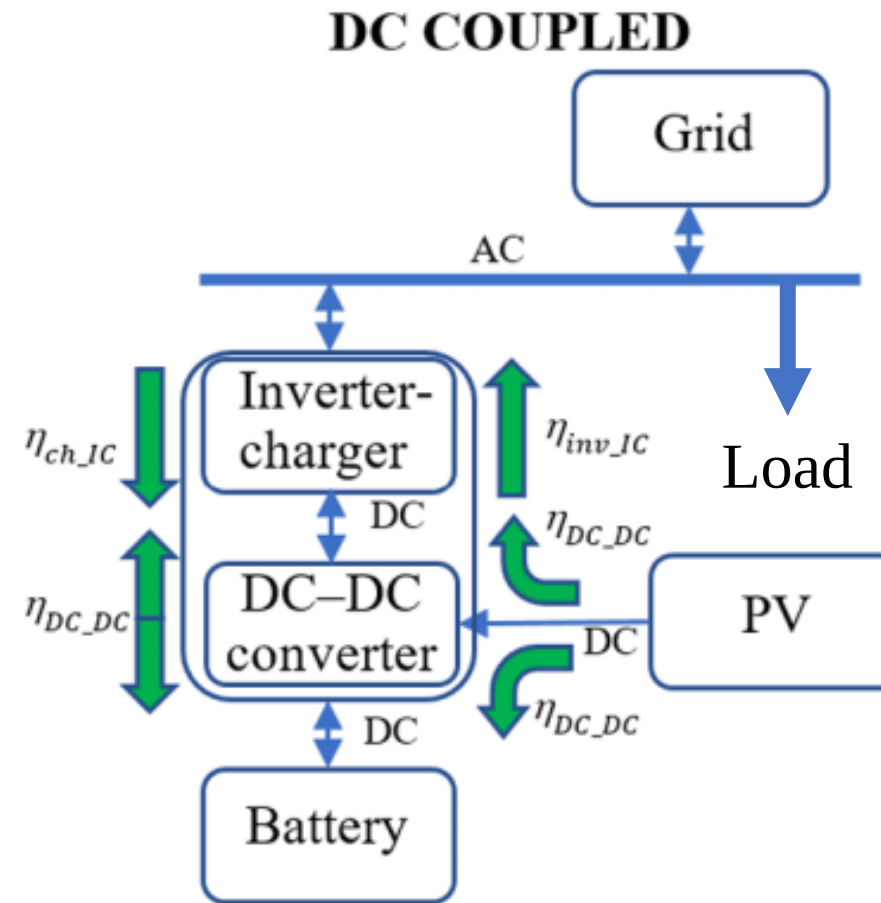
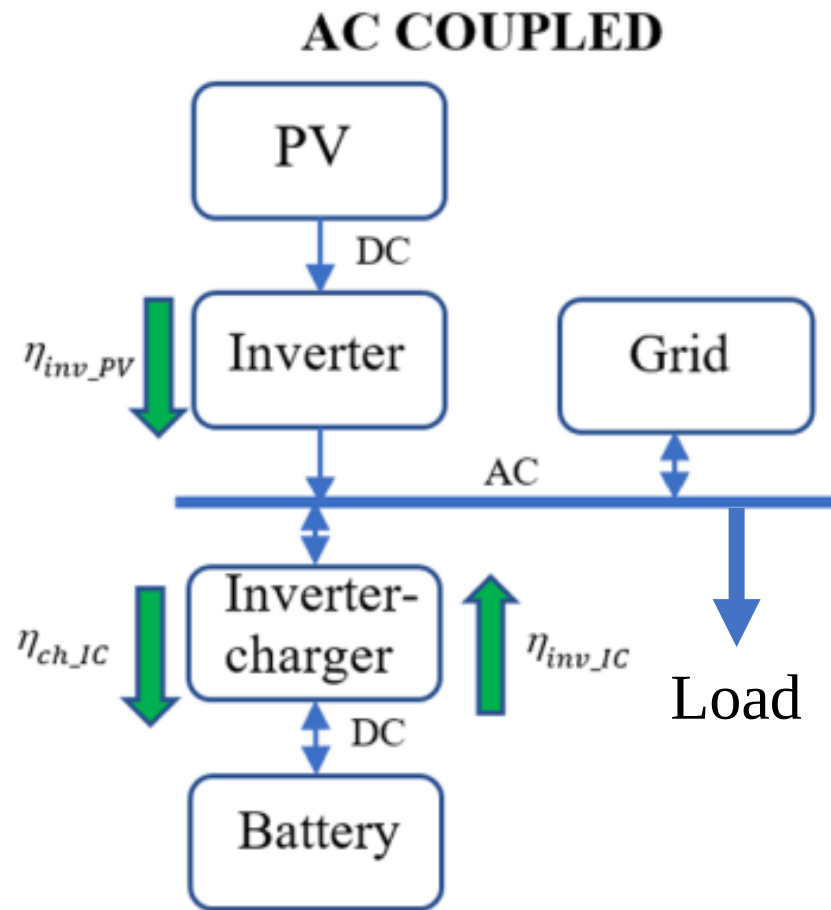


Construyendo modelos de despacho de AGPE



- Caracterización de la demanda
- Caracterización del mercado eléctrico
- Requerimientos regulatorios (CREG 174-2021, Colombia)
- Caracterización del recurso solar (pasado, futuro)
- Pre-Dimensionamiento del sistema PV (AGPE < 1000 kW)
- Pre-Dimensionamiento del sistema BESS

Modelo de Despacho óptimo (Forecast 24horas)



Microgrid
Energy Management System
MgEMS

Dufo-López, R., Lujano-Rojas, J. M., Artal-Sevil, J. S., & Bernal-Agustín, J. L. (2024). Optimising Grid-Connected PV-Battery Systems for Energy Arbitrage and Frequency Containment Reserve. *Batteries*, 10(12), 427.

<https://www.mdpi.com/2313-0105/10/12/427MDPI+2Preprints+2ResearchGate+2>



Review

Energy Management Systems for Microgrids: Main Existing Trends in Centralized Control Architectures

Danny Espín-Sarzosa ¹, Rodrigo Palma-Behnke ^{1,*} and Oscar Núñez-Mata ² 

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² School of Electrical Engineering, University of Costa Rica, San José 11501, Costa Rica; oscar.nunezmata@ucr.ac.cr

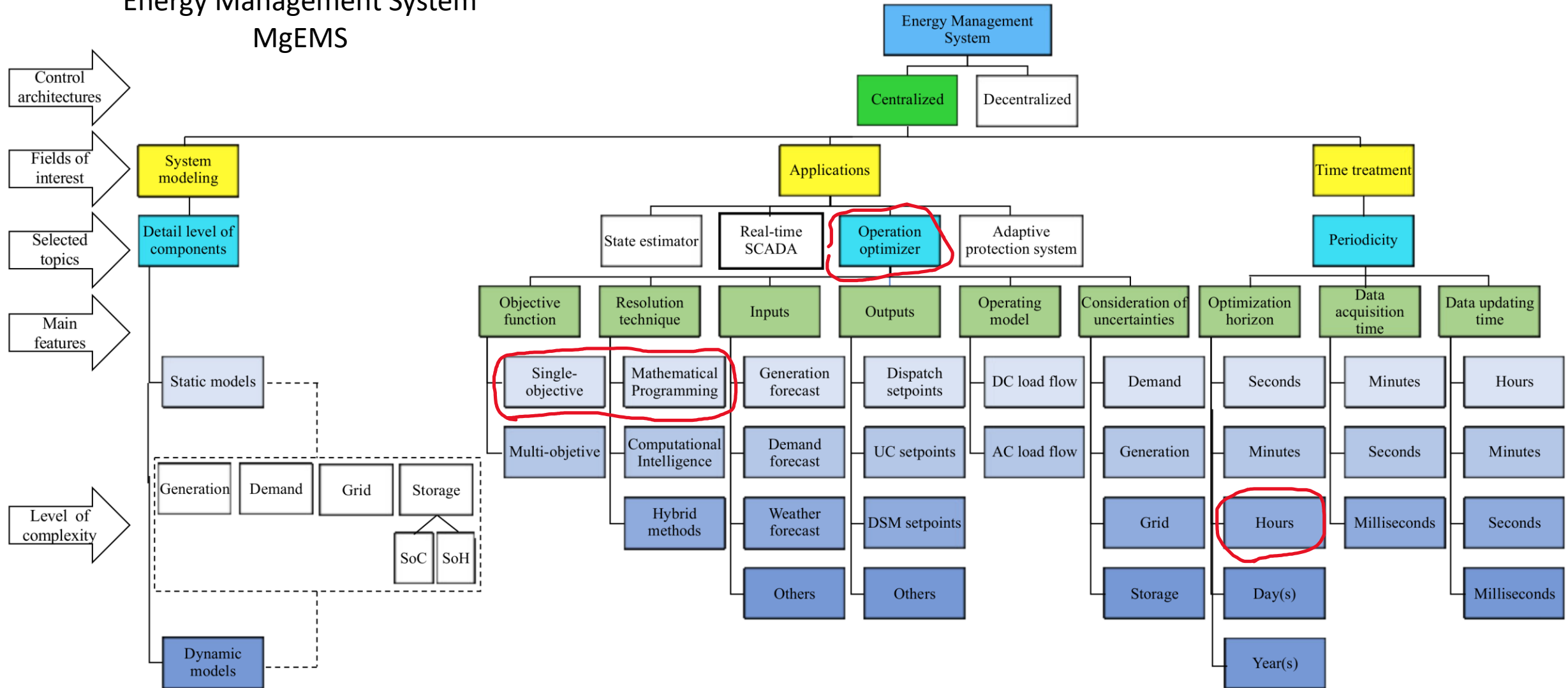
* Correspondence: rodpalma@cec.uchile.cl; Tel.: +56-9576-96810

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Espín-Sarzosa, D., Palma-Behnke, R., & Núñez-Mata, O. (2020). Energy management systems for microgrids: Main existing trends in centralized control architectures. *Energies*, 13(3), 547, <https://www.mdpi.com/1996-1073/13/3/547>

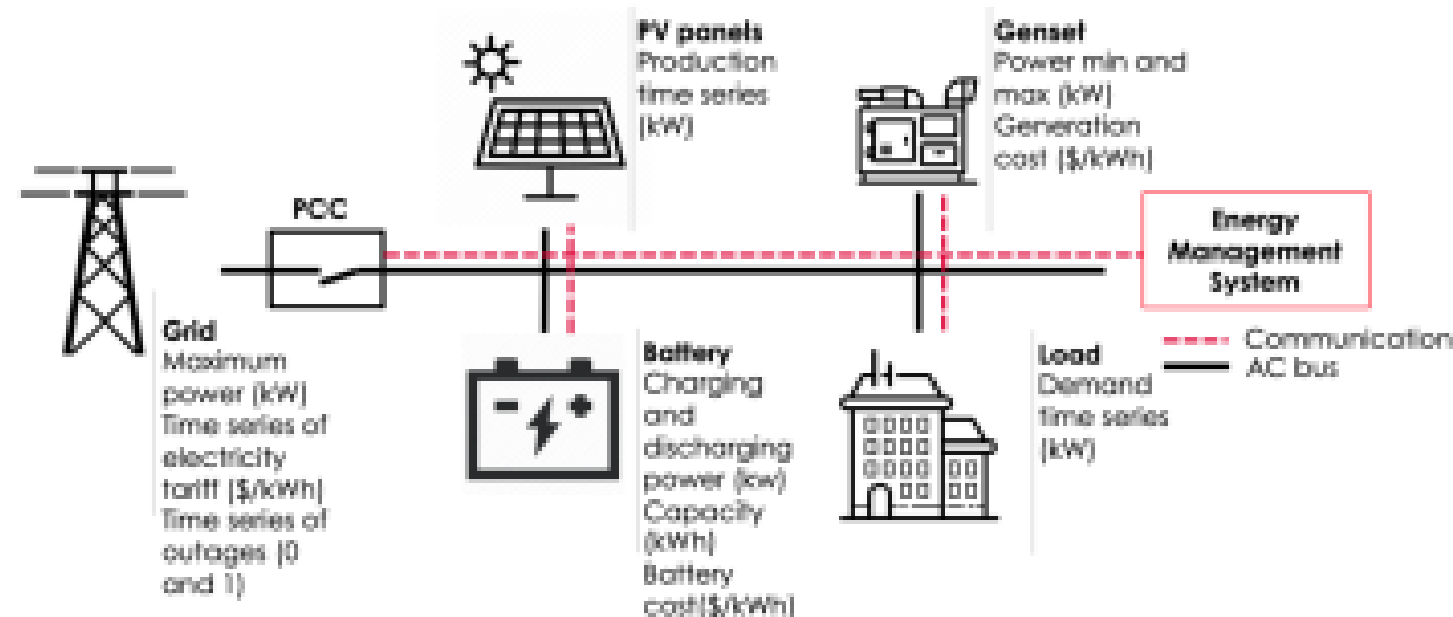
Microgrid Energy Management System MgEMS



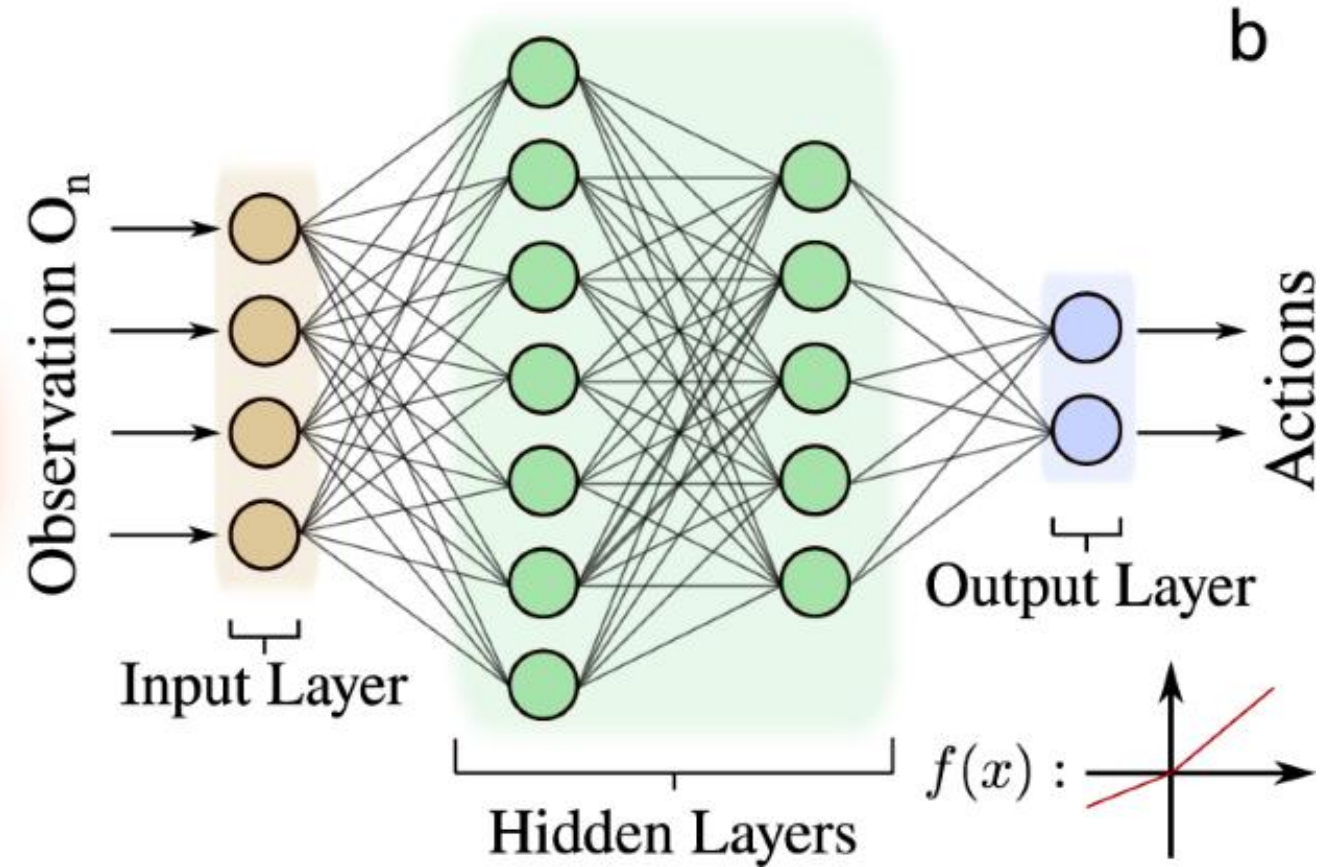
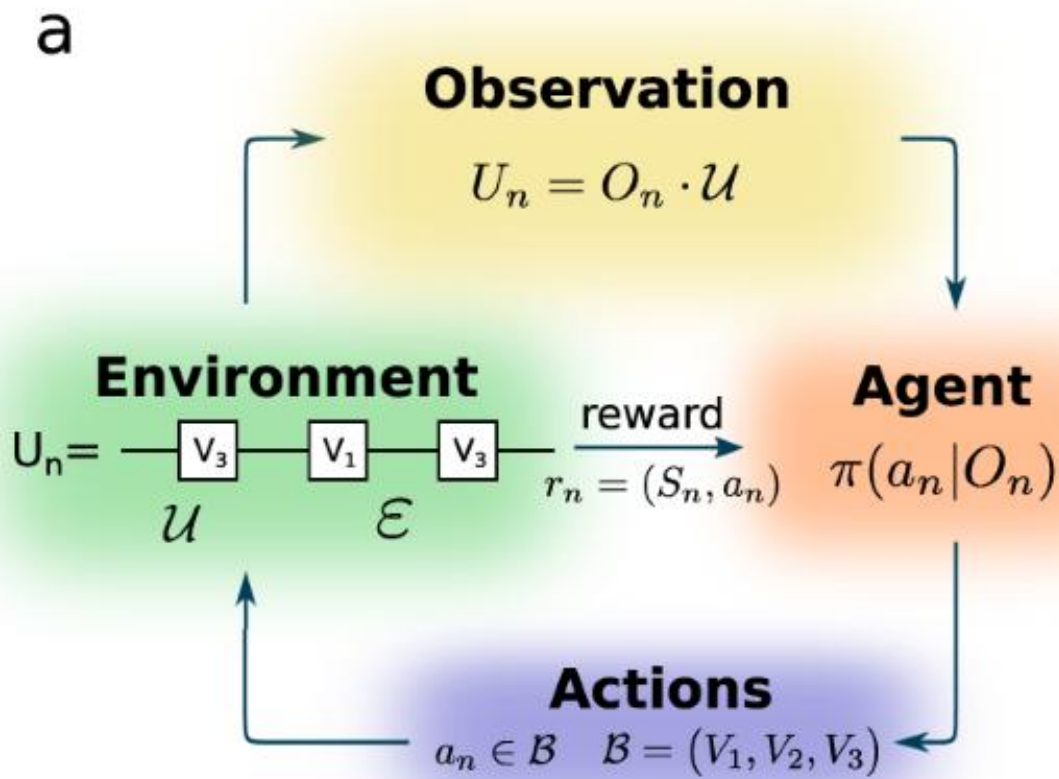
Espín-Sarzosa, D., Palma-Behnke, R., & Núñez-Mata, O. (2020). Energy management systems for microgrids: Main existing trends in centralized control architectures. *Energies*, 13(3), 547, <https://www.mdpi.com/1996-1073/13/3/547>

Energy Management System for Microgrids based on Deep Reinforcement Learning

César Garrido, Fernando Lozano, Guillermo Jiménez-Estévez, Luis G. Marín, and Carolina Higuera,



Garrido, César, et al. "Energy Management System for Microgrids based on Deep Reinforcement Learning." *2021 IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*. IEEE, 2021



Artificial Neural Networks



Article

Optimal Power Dispatch for Maximum Energy Community Welfare by Considering Closed Distribution Systems and Renewable Sources

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² Department of Electrical Engineering, University of Zaragoza, María de Luna 3, 50018 Zaragoza, Spain

* Correspondence: jmyusta@unizar.es

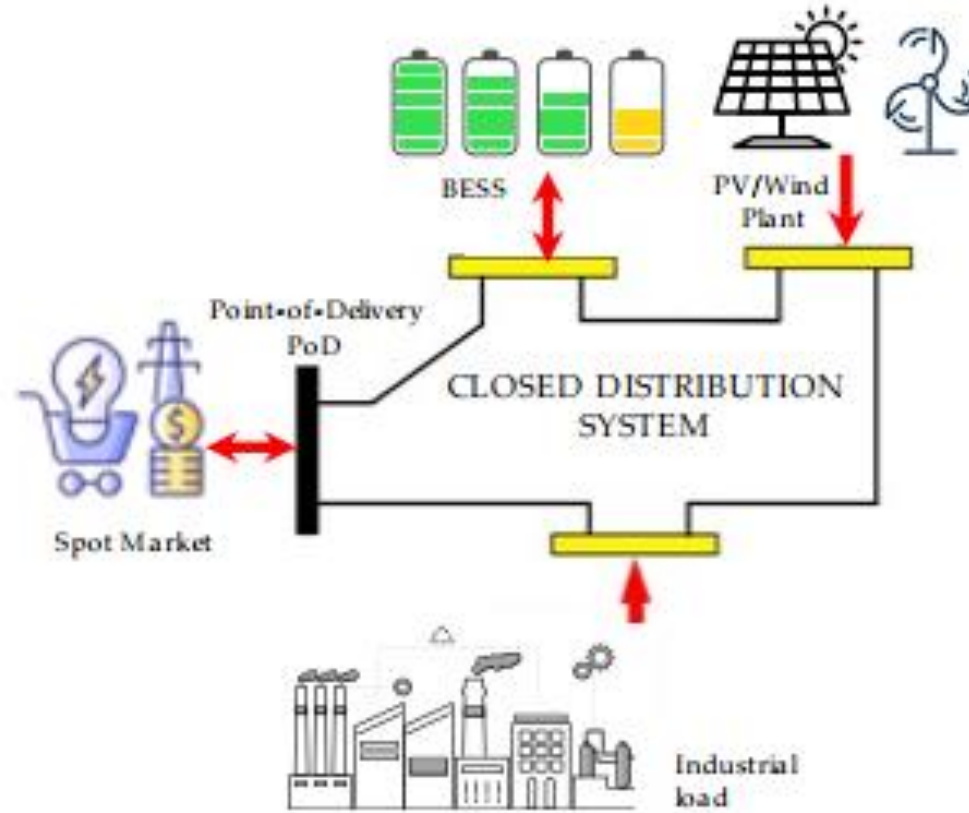


Figure 1. Energy community whose CDS is connected to the public grid with a single metering frontier or point of delivery.

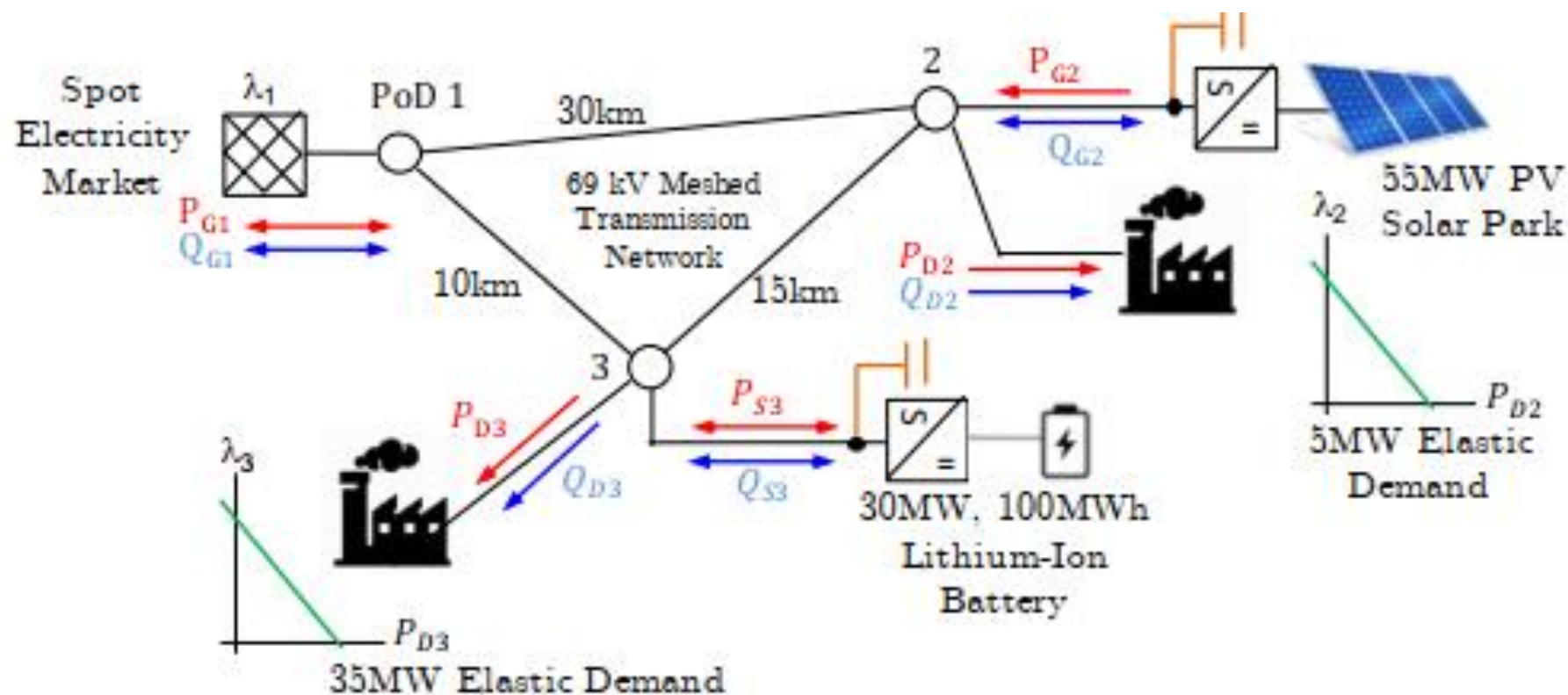


Figure 5. Test system: EC with two industries, with PV and BESS assets connected to a CDS.

Algorithm 1 Community Welfare Optimal Power Dispatch

Input: Set up the case-study economic/technical data

- 1: Define the objectives of the model
- 2: Define the PV/Wind generation model
- 3: Define the Load model
- 4: Define the BESS model
- 5: Define the CDS model
- 6: Scripting the optimization problem
- 7: Solve the optimization problem
- 8: Plot the energy community prices
- 9: Plot the active/reactive power dispatch
- 10: Print the energy balance
- 11: Print the economic balance (OPEX results)
- 12: Define CAPEX requirements
- 13: Determine financial indexes

Output: Print results: NPV, IRR, PBT, B/C ratio.
